

Tokenized Asset & Fractional Ownership Platform

Tokenize a real-world asset with KYC-gated, compliance-aware fractional ownership.

DIFFICULTY MEDIUM

What the team builds

Pick one real-world asset class — gold, an invoice, or fractional real estate — and represent it on-chain as a token that can be split into fractions and transferred only between approved (whitelisted) holders. This extends the Gold-Token lab from the syllabus into something that resembles a real Real-World-Asset (RWA) product, where the genuinely hard question is not the code but what the token actually represents off-chain and who is legally allowed to hold it. That makes it an ideal project for a team where the domain expert can own the compliance and custody story.

Curriculum it draws on

- **FinTech course:** Digital Identity, Data in Financial Services, and the RegTech / compliance sessions.
- **Programming course:** ERC-20 / ERC-1155 token standards (S12–14), the Gold-Token lab (S14), access control & whitelisting (S15), and optionally The Graph to render holdings (S20–21).

How the work splits across the team

TECHNICAL LEAD · the builder

- Implement the asset token (ERC-20 for fractions, or ERC-1155).
- Build the mint / issue flow against a defined supply.
- Add a KYC whitelist gate so only approved addresses can receive.
- Optionally index holdings with The Graph for the UI.
- Tests + Slither pass on the transfer-restriction logic.

DOMAIN LEAD · non-technical

- Choose the asset class and justify it.
- Define what one token legally / economically represents.
- Design the custody and off-chain backing model.
- Build the KYC / eligibility & compliance rules.
- Own the valuation logic and the investor-facing pitch.

The compliance, custody and 'what does the token actually represent' workstream is the most interesting part of any RWA product — and it is almost entirely a domain-expert job, which makes this a strong balanced pairing.

Scope for one month

MVP (must finish)	Stretch (if ahead)
• Mint a tokenized asset with a fixed supply. • Transfer fractions between whitelisted holders only. • Block transfers to non-approved addresses.	• The Graph subgraph to show the holder registry. • A simple dividend / yield distribution. • Role-based admin for the whitelist.

Is this the right project for your team?

PICK THIS PROJECT IF...

- Your builder is comfortable with token standards and wants to go one step beyond a plain ERC-20.
- Your non-technical member is interested in compliance, custody, valuation, or asset structuring and wants a meaty analytical role.
- The team wants a project that looks and feels like a **real fintech product**, with a clear regulatory dimension.

Shared deliverables for every project: contract deployed & verified on the Sepolia testnet · thin React/HTML frontend (Sessions 20–21) · Hardhat or Foundry test suite · one Slither static-analysis pass (Session 16) · a short design document and a live demo. Teams pair one technical and one non-technical member; both names appear on every deliverable.